

The Covariance Audit: Why the Native Deformation Cannot Ride the Boosts, and Where Interactions Must Enter

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Abstract. The interaction era's central open risk, named at its opening and carried through Steps 1 and 2, is audited here and resolved — with a verdict sharper than either anticipated outcome. The tower's native interaction candidate is the first-order remnant of the second contraction, the R-flux deformation $[X^i, X^j] = i\mu \varepsilon^{ijk} P_k$. The audit establishes four facts, each machine-checked. (1) The dilation survives the deformation on the generators — the would-be correction to $[D, X^i]$ vanishes by ε -symmetry (verified at 0.0) — but the *relations* are scale-covariant only if μ itself transforms with weight three: the deformation converts the exact dilation automorphism into an isomorphism flow along the μ -axis. The tower's coupling is dimensionful — it is the R-flux density $R^{ijk} = \mu \varepsilon^{ijk}$ — and scale symmetry becomes a renormalization-group motion on it. (2) **The categorical exclusion:** the Jacobiator of the deformed bracket is nonzero at first order, $[X^1, X^2, X^3]_{\text{Jac}} = 3\mu\hbar$ — this is what "dies linearly" meant — and by the series' founding no-representation theorems, non-associative brackets admit no Hilbert-space operator realization. At any $\mu \neq 0$ the deformation cannot surface as an operator deformation of the assembled net. (3) The obstruction is geometric and classical too: the flux two-form has $dF \neq 0$ (the R-flux three-form, $-3/\hbar$), so no redefinition of coordinates restores commutativity even at the Poisson level. (4) **Frame anchoring:** no constant three-index tensor is Lorentz invariant — the boost action on the embedded ε^{ijk} is nonzero (norm 2.45, machine-checked) — so even in non-operator, phase-space formulations of the deformation, the ε structure selects the fiber's rest frame and cannot ride the assembled boosts. The exact-or-absent dichotomy of the series' Lorentz audit is thereby resolved on *absent*: the emergent theory's Lorentz invariance is safe categorically, not by smallness of a coupling. The constructive conclusion maps where interactions must enter instead: the state stratum (realized by the equation of state), source-type couplings of representable currents (located by the measurement gate as the sector doorway), and the sedenion-level multiplicity structure as a source of coupling patterns and selection rules rather than deformed kinematics. With this, the era's three named steps are complete.

1. The question

Since the era analysis, one risk has stood above the others (H2): the tower's own finite- μ remnants — the project's native candidate for interaction structure — were born in a graded frame, and it was unknown whether they could be carried covariantly on the assembled, exactly Lorentz-invariant theory, or would break it, or would be confined. The audit decides.

2. The deformation, and what it does to scale

The first-order remnant below the second contraction's endpoint is

$$[X^i, X^j] = i\mu \varepsilon^{ijk} P_k, \quad [X^i, P_j] = i\hbar \delta_j^i, \quad [P_i, P_j] = 0.$$

****Proposition 1.**** The dilation generator $D = \frac{1}{2}(X \cdot P + P \cdot X)$ acts on the generators exactly as at $\mu = 0$: the deformation's contribution to $[D, X^i]$ is $\varepsilon^{kij}(P_j P_k + P_k P_j)/2 = 0$ by antisymmetry against symmetry — verified at 0.0. But the *relation* $[X, X] = i\mu\varepsilon P$ carries dilation weight mismatch (-2) against $(+1)$: it is covariant only under $\mu \mapsto \lambda^3 \mu$. Hence at $\mu \neq 0$ the dilations cease to be automorphisms of one algebra and become isomorphisms along the μ -family. ■

Read physically: μ is a dimensionful coupling — the R-flux density $R^{ijk} = \mu\varepsilon^{ijk}$, weight three, a volume scale — and the scale symmetry that powered the entire bridge becomes, at $\mu \neq 0$, a renormalization-group motion on the coupling. The tower thus supplies, by itself, the structure of a dimensionful interaction; whether it can *enter the operator theory* is the next question, and the answer is no, twice over.

3. Theorem 1: the categorical exclusion

Theorem. At any $\mu \neq 0$ the deformed brackets admit no Hilbert-space operator representation, and therefore cannot surface as a deformation of the assembled net's observables.

Proof. The Jacobiator is first order: $[[X^1, X^2], X^3] + \text{cyc.} = i\mu\varepsilon^{12l}[P_l, X^3] + \text{cyc.} = 3\mu\hbar$ — nonzero (displayed arithmetic). Operator commutators on a Hilbert space satisfy the Jacobi identity identically; the series' founding no-representation theorems make exactly this obstruction precise for the octonionic bracket family. Hence no operator realization exists for $\mu \neq 0$. ■

The obstruction is not removable by cleverness at the classical level either: seeking new coordinates $X' = X + O(\mu)$ that commute requires the flux two-form $F_{ij} = -\varepsilon_{ijk}P^k/\hbar$ to be exact, but dF is the R-flux three-form, $(dF)_{123} = -3/\hbar \neq 0$ (displayed): the twist is cohomologically nontrivial. "Dies linearly" was always the precise statement: the Jacobiator vanishes *with* μ , and only at the contraction's endpoint — the associative, representable world — does the operator theory begin.

4. Theorem 2: frame anchoring

Theorem. Even independently of associativity, the deformation cannot be Lorentz covariant: there exists no constant three-index tensor invariant under $SO(3, 1)$, so any fixed R^{ijk} selects a preferred frame.

Proof and check. The invariant tensors of the Lorentz group are generated by $g_{\mu\nu}$ and $\varepsilon^{\mu\nu\rho\sigma}$ — even and four-index; no three-index invariant exists. Machine check: embedding ε^{ijk} as the spatial components of a 4-tensor and acting with a boost generator gives $\|\delta_{\text{boost}}T\| = 2.45 \neq 0$. The spatial ε is $\varepsilon^{0ijk}u_0$ in disguise: the deformation carries the fiber's rest frame within it. ■

This second exclusion matters because the deformation *does* possess a formalism of its own — the non-associative phase-space and star-product framework of the R-flux literature (Günaydin–Lüst–Malek). Within that non-operator setting the structure is consistent;

Theorem 2 says that even there it is frame-anchored, confined below the geometric assembly. Two independent walls: associativity and covariance.

5. The verdict

The exact-or-absent dichotomy of the series' Lorentz audit is resolved on **absent** — and absent in the strong sense: not tuned away, not suppressed, but categorically excluded from the representable stratum by theorems the series proved before it knew it would need them here. The emergent theory's Lorentz invariance is safe by structure. The μ -stratum's honest role is thereby fixed as what the tower's papers always described: the pre-geometric, pre-associative origin story — the layer where $\hbar$ is forged and the scale line is born — expressible in its own phase-space language, and never as operator interactions of the world it produces.

6. Where interactions must enter

The audit's constructive face is a map with three roads, all already open. **(i) The state stratum:** interactions of the geometry — realized: the equation-of-state derivation shows gravity entering through entropy balance, with the metric a thermodynamic variable; couplings living in states and modular data face no exclusion. **(ii) Source-type couplings of representable currents:** the measurement gate proved bilinear compact couplings exist, are causal, and preserve the memory sector, and located sector dynamics precisely at linear, "charged" couplings — the doorway for matter interactions, radiation, and memory physics, all inside the operator theory. **(iii) The multiplicity structure:** the sedenion-level generation algebras — the complement's surviving physical employment per the revised opening paper — can supply *coupling patterns and selection rules* (which currents couple to which, with what relative strengths) without deforming kinematics: octonionic data as the pattern of interactions, not their mechanism. The forward program of the era is model-building within these three channels, and nothing in the audit obstructs it.

7. Scope

Five items. *(i)* The audit covers the first-order remnant; every finite- μ member of the contraction ladder short of the endpoint carries a nonzero Jacobiator by the same arithmetic, so the exclusion is ladder-wide. *(ii)* The phase-space formalism of the R-flux tradition is the μ -stratum's own language and is cited, not adjudicated. *(iii)* Whether μ -structure imprints on the *values* of couplings and state-stratum constants — an imprint on numbers rather than on kinematics — is left open and flagged as the interesting residue. *(iv)* The experimental side of the original Lorentz audit (the radiative-sensitivity constraints) is untouched and now moot for this channel: nothing surfaces to be constrained. *(v)* Era status: Steps 1, 2, and 3 are complete; the interaction era's foundations are closed, and what follows is construction within the allowed channels.

References

- Companion papers: *The Second Contraction: From Algebraic R-Flux to the Heisenberg Algebra, the Dilation Symmetry It Creates, and the SO(3) Remnant of G₂; The Central Element of Algebraic R-Flux; The Assembly: Null Lines Through a Point,*

the Light-Cone Identification, and Lorentz Symmetry as Output; The Measurement Gate: The Probe Scheme on the Net, Unsharp Induced Observables, and Sector-Preserving Scattering; The Equilibrium Constraint: One Killing Vector's Two Faces, the Geometric Lemma, and the Linearized Einstein Equation as an Equation of State.

- Günaydin, M., Lüst, D. & Malek, E., "Non-associativity in non-geometric string and M-theory backgrounds, the algebra of octonions, and missing momentum modes," arXiv:1607.06474.
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